Vendor	Product	Description of MAID features
COPAN Systems	Revolution 300A, 300M, 300T/TX	Only 25% of drives can be spun up at any one time, maximizing energy savings.
EMC	Disk Library 4000 series	Spindown automatically places inactive drives in sleep mode and activates them on demand, but they must be set by disk drawer and not individual drives.
Fujitsu	(in ECO mode) ETERNUS8000, ETERNUS4000, ETERNUS2000	ECO mode manages the on/off state of the disk drives' spindle motors based on policy settings as well as usage patterns. This means the disk spins only during a set period unless the disk is accessed more often than expected.
HDS	AMS 1000, 500, 200; WMS 100	HDS is a professional services engagement to select RAID groups appropriate for power savings and customize scripts to spin up and spin down drives.
NEC	D1-10, D3-10, D8-10	Using dedicated software control, this system turns off the motor power of unused disk drives. This leads to power consumption reductions of up to 30% compared with conventional models.
Nexsan Technologies	SATABeast, SATABeast Xi, SATABoy, Assureon, DATABeast, SASBoy	AutoMAID is user-selectable, enabling users to determine the level of response time performance to energy savings. Supports VTL, full file format D2D, and standard RAID implementations.

Source: Forrester Research

that is powered down, power management software will spin down an operational LUN when it's no longer in use and then power up the LUN that contains the data being requested. This means a delay in getting data on or off LUNs depending on what drives are currently spinning at the time of the request. The reduction in power consumption can be as much as 85percent compared with traditional storage arrays in which all disks are spinning, but not all applications can tolerate the latency introduced by the spin-up process.

Lack of software tools to identify appropriate data for MAID. Most organizations have little to no insight into how much or what kind of data they have. Couple this with a shortage of software to specifically identify persistent data suitable for MAID storage and it's no surprise adoption has been slow. Most storage architects know that transactional data on a MAID system will fail, but beyond that, they're not confident about which of their data has the right access patterns for MAID. It would be helpful if archiving software vendors identified which data within their applications goes on what storage device and moved it there, but so far vendors have been slow to do this work.

Short refresh cycles. Tech refresh cycles have been extended, but not by much. One of the toughest problems in building a long-term archive is preserving the data in the system through multiple generations of technology. COPAN alleges it has been able to extend the life of its storage from the typical three-to-five-year upgrade cycle to seven years. It's a small step forward when you consider building an archive that will last a hundred years.

MAID Vendors

Vendors with MAID storage products include COPAN Systems, EMC, Fujitsu, Hitachi Data Systems (HDS), NEC, and Nexsan Technologies

The Future

Although the lack of ways to consume MAID storage has delayed its adoption, that's set to change, as the biggest proponents of the technology, COPAN and Nexsan, have recently introduced network-attached storage (NAS) technologies to branch out beyond the backup market. COPAN has bundled Quantum's StorNext file system with its MAID arrays to enable file archiving, and Nexsan recently unveiled a NAS gateway based on Windows Storage Server. Furthermore, the combination of data deduplication and replication will mean that vaulting data offsite on disk is a possibility. MAID will enable data vaulting companies that use low-grade data center space in terms of power and cooling to offer a disk-based backup, restore, and vaulting service.

MAID is also evolving from a binary “on” or “off” mode to a more flexible power management scheme that balances energy consumption against the performance and availability needs of the application.

On the issue of energy efficiency, data center power and cooling considerations are set to become more prevalent, especially in Europe and the Far East where this is already a critical problem.

Vendors need to reduce the power consumption of all components in the system, not just the drives, according to Ethan Miller, professor of computer science at the University of California in Santa Cruz. He says the controllers and CPUs in MAID storage technology can consume 150 to 200 watts apiece and are constantly running. Miller is working on a project in archival storage called Pergamum, which aims to develop a long-term archive using low-power processors, SATA disks, and non-volatile random access memory (NVRAM) such as flash to hold metadata stores. When you consider the implications of searching an exabyte of storage, the address tables for all that data better be on fast storage.

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